ControlShield: Network Security and Forensics for Industrial Control Systems

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***Abstract*—Industrial Control Systems (ICS) are critical components of modern automated infrastructure, encompassing sectors such as logistics, manufacturing, energy and utilities. Security and forensics on Programmable Logic Controllers (PLC) are paramount due to the potential consequences of cyber threats and vulnerabilities. This project presents ControlShield, an Industrial PLC Guardian that provides integrated intrusion detection and prevention capabilities to safeguard ICS environments.**

The experiments conduct greybox analysis on a simulated ICS network with a TM212CE24T PLC, HMIS5T HMI and a managed TP-Link TL-SG105E switch. The PLC Guardian is a Windows based python application that monitors the network for malicious activity and provides real-time alerts and mitigation actions. The design theory is a fully integrated PLC IPS that communcates directly with the PLC to prevent malicious activity.

The results demonstrated the effectiveness of ControlShield in near real-time detection using port mirroring and packet analysis. The PLC Guardian successfully identified and mitigated various attack scenarios, including exploitation of exposed services such as diagnostics web portal and Modbus TCP attacks. There is also experiments conducted to blackbox footprinting and reconnaissance attacks for system discovery.

In conclusion, ControlShield provides a comprehensive solution for enhancing the security and resilience of Industrial Control Systems.

***Index Terms*—Industrial control systems, programmable logic controllers, network security, intrusion detection and prevention, network forensics, cybersecurity.**

I. Introduction

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mds
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A. System Design

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1) ICS Network Architecture: Subsubsection text here.

2) PLC Guardian Design: Subsubsection text here.

B. Experiments and Results

Subsection text here.

1) Baseline Network Traffic Analysis: Subsubsection text here.

2) Exposed Services Exploitation: Subsubsection text here.

3) Modbus TCP Attack: Subsubsection text here.

4) Intrusion Prevention and Mitigation: Subsubsection text here.

II. Conclusion

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Appendix A
Proof of the First Zonklar Equation

Appendix one text goes here.

Appendix B

Appendix two text goes here.

The authors would like to thank...

References

[1] H. Kopka and P. W. Daly, *A Guide to LaTeX*, 3rd ed. Harlow, England: Addison-Wesley, 1999.

The authors are students at California Polytechnic Pomona and are not IEEE members.



Michael Shell Biography text here.

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